

DIFFERENTIATION WITH THE PRODUCT RULE: PROBLEM SHEET

Complete the problems below. Solutions and marking available on request.

Recall that if $y = u \cdot v$ then

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}.$$

1. Find the derivative dy/dx for the following functions.

- (a) $y = 2x \sin(x)$,
- (b) $y = (7x^3 - 12x + 4) \cos(x)$,
- (c) $y = 10^x x^2$,
- (d) $y = \sin(x) \tan(x)$,
- (e) $y = \ln(x) \cos(x)$.

2. Find the derivative dy/dx for the following functions.

- (a) $y = x^3 e^x \sin(x)$,
- (b) $y = (6x - 2) \cos(x) \tan(x)$.

The following problems may also include the chain rule.

3. Find the derivative dy/dx for the following functions.

- (a) $y = \ln(x \sin(x) \cos(x))$,
- (b) $y = e^{x^2-1} \tan(5xe^x)$,
- (c) $y = (\sqrt{x} + 4)e^{\sin(x)}$.

4. Let $P(x) = f(x)g(x)$ and $a \in \mathbb{R}$. Show that if $f(a) = 0$ and $f'(a) \neq 0$, then $P'(a) = 0$ if and only if $g(a) = 0$.

(‘If and only if’ says both (i) $P'(a) = 0$ means $g(a) = 0$, and (ii) $g(a) = 0$ means $P'(a) = 0$)

5. Differentiate $f(x) = \cos(x \sin(x \cos(x)))$.

6. Use the product rule to explain why, for all $C \in \mathbb{R}$,

$$\frac{d}{dx}(Cf(x)) = Cf'(x).$$

7. Let $W(x) = f(x)g'(x) - f'(x)g(x)$. Find the derivative $W'(x)$.

(This is the ‘Wronskian’ of two functions f and g)